

Implementing Fairness Constraints in Spectral Clustering

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Abstract

Machine learning algorithms are known to amplify small biases present in their training data, which raises concerns about the equity of clustering methods used to partition real-world data. In this paper, we study spectral clustering, a graph partitioning method based on the eigenvectors of a graph’s Laplacian matrix, and investigate how an explicit fairness constraint can be incorporated into its optimization problem. We define fairness in terms of *balance*, a measure of how proportionally different subgroups are represented within each cluster, and reformulate the spectral clustering objective to enforce this constraint directly. We further propose a relaxation of the orthogonality constraint on the clustering matrix, replacing it with a linear constraint, in an effort to reduce the computational cost of the eigendecomposition step. We evaluate both the fairness-constrained and unconstrained clustering methods on two real-world social network datasets, FriendshipNet and FacebookNet, and compare their balance scores across a range of cluster counts.

1 Introduction

The recent growth in machine learning research has led to the widespread adoption of algorithms whose behavior is difficult to audit for fairness. Many of these algorithms have been shown to amplify small biases already present in their training data, raising questions about the equity and ethics of their use. A well-known example is Microsoft’s Tay chatbot, which was taken offline in 2016 after producing racist remarks as a result of being trained on unmoderated Twitter data [1].

In this paper, we study *spectral clustering*, a machine learning method for partitioning graph data based on a similarity or affinity matrix. Spectral clustering groups data points by examining the eigenvectors of a graph’s Laplacian matrix together with normalized cuts, forming communities of similar nodes [6]. Our goal is to identify a mathematical notion of fairness for this setting and to incorporate it directly into the clustering algorithm, rather than treating fairness as a post-hoc correction.

2 Background

2.1 Normalized Cut and Spectral Clustering

Given a graph with vertex set V partitioned into two subgraphs A and B , the *normalized cut* (Ncut) and *ratio cut* (RatioCut) between A and B are defined as

$$\text{Ncut}(A, B) = \frac{\text{cut}(A, B)}{\text{assoc}(A, V)} + \frac{\text{cut}(A, B)}{\text{assoc}(B, V)}, \quad (1)$$

$$\text{RatioCut}(A, B) = \frac{\text{cut}(A, B)}{|A|} + \frac{\text{cut}(A, B)}{|B|}, \quad (2)$$

where $\text{cut}(A, B)$ denotes the total weight of edges crossing between A and B , and $\text{assoc}(A, V)$ denotes the total weight of edges with at least one endpoint in A . Note that $\text{assoc}(A, V) \approx |A|$ and $\text{assoc}(B, V) \approx |B|$.

These definitions extend naturally to a k -way partition $A_1, \dots, A_k$:

$$\text{Ncut}_k = \sum_{i=1}^k \frac{\text{cut}(A_i, V - A_i)}{\text{assoc}(A_i, V)}, \quad \text{RatioCut}_k = \sum_{i=1}^k \frac{\text{cut}(A_i, V - A_i)}{|A_i|}. \quad (3)$$

Solving these partitioning problems exactly is computationally intractable for graphs of any meaningful size, since it requires searching over all possible discrete partitions of the vertex set. Spectral clustering relaxes this discrete optimization problem into a continuous one that can be solved using the eigenvectors of the graph Laplacian $L = D - W$, where W is the affinity matrix of the graph and D is the diagonal degree matrix with $d_i = \sum_{j=1}^n W_{ij}$.

2.2 Defining Fairness via Balance

Suppose the data set V contains h subgroups such that $V = V_1 \cup \dots \cup V_h$, and suppose a clustering of V into k clusters is given by $C_1, \dots, C_k$, so that $V = C_1 \cup \dots \cup C_k$. Following the fairness notion of Chierichetti et al. [2], we define the *balance* of a cluster C_l as

$$\text{balance}(C_l) = \min_{s \neq s' \in [h]} \frac{|V_s \cap C_l|}{|V_{s'} \cap C_l|} \in [0, 1]. \quad (4)$$

A cluster has balance close to 1 when every subgroup is represented within it in roughly equal proportion, and balance close to 0 when the cluster is dominated by a single subgroup. A clustering is considered *statistically fair* if, for every cluster C_l and every subgroup V_s ,

$$\frac{|V_s \cap C_l|}{|C_l|} = \frac{|V_s|}{|V|}, \quad (5)$$

meaning each subgroup is represented in every cluster in the same proportion as in the overall data set.

3 Methods

3.1 Reformulating RatioCut

To solve the RatioCut minimization problem, we represent a clustering using a matrix $H \in \mathbb{R}^{n \times k}$ with entries

$$H_{il} = \begin{cases} 1/\sqrt{|C_l|}, & i \in C_l, \\ 0, & i \notin C_l, \end{cases} \quad (6)$$

which allows the RatioCut objective to be written compactly as

$$\text{RatioCut}(C_1, \dots, C_k) = \text{Tr}(H^T L H) \quad \text{subject to } H \text{ of the form above.} \quad (7)$$

Because the discrete set of matrices H satisfying this exact form is extremely limited, computing the optimal clustering directly is time-consuming. It is therefore standard to relax this constraint to the orthogonality condition $H^T H = I_n$, which is far less restrictive and significantly reduces computation time. The tradeoff is that H no longer directly represents a discrete clustering once relaxed, so an additional discretization step (such as k -means on the rows of H) is required to recover an explicit partition.

3.2 Incorporating a Fairness Constraint

To enforce fairness within the clustering, we define a matrix $F \in \mathbb{R}^{n \times (h-1)}$ whose columns are the vectors

$$f^{(s)} = \left(\frac{|V_s|}{n} \right) \cdot \mathbf{1}_n, \quad s \in [h-1], \quad (8)$$

where $f^{(s)}$ is the group-indicator vector for subgroup V_s . The matrix F is constructed directly from the balance definition in Equation (4) together with the structural properties of H under the relaxed orthogonality constraint. The resulting fairness constraint is

$$F^T H = 0_{(h-1) \times k}, \quad (9)$$

which restricts the clustering matrix H to lie in the null space of F^T , guaranteeing that every cluster's subgroup composition reflects the subgroup composition of the full data set.

Combining this with the relaxed RatioCut objective gives the fairness-constrained optimization problem

$$\min_{H \in \mathbb{R}^{n \times k}} \text{Tr}(H^T L H) \quad \text{subject to } H^T H = I \text{ and } F^T H = 0_{(h-1) \times k}. \quad (10)$$

3.3 Eliminating the Eigendecomposition

The orthogonality constraint $H^T H = I$ requires solving a generalized eigenvalue problem, which remains a computational bottleneck for large graphs. We investigate whether this step can be avoided entirely by replacing the orthogonality constraint with a linear constraint of the form $BH = C$, giving the modified problem

$$\min_{H \in \mathbb{R}^{n \times k}} \text{Tr}(H^T L H) \quad \text{subject to } BH = C \text{ and } F^T H = 0_{(h-1) \times k}. \quad (11)$$

Replacing an eigendecomposition with a linear system is, in principle, far less computationally expensive. Our results below evaluate whether this substitution preserves comparable clustering quality, as measured by balance, relative to the original eigenvector-based formulation.

4 Results

We evaluated four clustering variants, unnormalized spectral clustering, normalized spectral clustering, fairness-constrained unnormalized spectral clustering, and fairness-constrained normalized spectral clustering, as well as a fairness-constrained variant of the linear-constraint reformulation (denoted FAIR CNCut), on two real-world social network datasets: FriendshipNet and FacebookNet. For each method, we measured the balance of the resulting clustering, as defined in Equation (4), across cluster counts $k = 2$ through $k = 8$.

Across both datasets, we observed the following consistent patterns:

- Balance decreases as the number of clusters k increases, for every method tested. This is expected, since partitioning a graph into more, smaller clusters makes it progressively harder for every subgroup to be proportionally represented within each cluster.
- The fairness-constrained methods generally achieved higher balance values than their unconstrained counterparts at the same value of k , indicating that the fairness constraint $F^T H = 0$ is successfully improving subgroup representation within clusters.

- Replacing the orthogonality constraint with the linear constraint $BH = C$ in the FAIR CNCut variant did not significantly change the resulting balance scores relative to the standard fairness-constrained spectral clustering methods, suggesting that the linear relaxation may be a viable, less computationally expensive alternative, though we did not directly measure wall-clock runtime in this study.

5 Applications

Fairness-constrained spectral clustering has a number of practical applications, including:

- Partitioning social networks into balanced friend groups or communities;
- Segmenting images based on pixel similarity while controlling for representational balance across regions;
- Tracking the occurrence of diseases or traits across a population in a way that avoids systematically under- or over-representing particular subgroups;
- Studying the differential effects of diseases across demographic groups.

6 Limitations and Future Work

This study has several limitations that motivate future work. First, while our results show that fairness-constrained methods achieve higher balance scores, we have not yet rigorously established the precise relationship between the balance metric and a complete, formal notion of fairness; this remains an open question. Second, we have not yet determined whether the linear constraint $BH = C$ yields a solution of comparable quality to the original eigenvector-based formulation across a wider range of graphs and parameter settings, beyond the balance comparisons reported here. Finally, we observed that the matrix F tends to be dense, which appears to negatively affect the efficiency of the algorithm; understanding and addressing the cause of this density is left to future work.

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